Sea level is likely to rise more than 1 meter in the 21st century

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Abstract

Every centimeter of sea-level rise exposes millions more people to annual coastal flooding, yet projections of sea-level are not yet reliable. Here, we construct a data-driven sea-level forecast by calibrating reduced physical models for individual sea-level contributors against decades-long independent observational records. When forced with observed surface temperatures, the calibrated components sum to match observed global mean sea level rise (GMSLR) within error. Our forecasts show a 73% probability of GMSLR exceeding 1 meter, and 10% of exceeding 2 meters, by the end of the century. The dynamics of West Antarctic glaciers account for $> 90\%$ of the forecast uncertainty and reveal orthogonal dimensions of sea-level risk: emissions reduction decreases median projections related to the predictable, temperature-sensitive components, but leaves the high-risk tail of the distribution largely unchanged, exposing hundreds of millions of people to accelerating sea-level risks.

Rates of global mean sea-level rise (GMSLR) have doubled twice in the past 50 years [1, 2] (Fig. 1). The first doubling took about 16 years and 0.34 °C of warming [3]. The second took about 21 years and 0.41 °C of warming (Fig. 1). The rate could double again around 2080 if the current observed sea-level acceleration continues, and by 2040 if it takes no more warming than the previous two doublings. The 40-year gap illustrates the uncertainty that coastal planning and adaptation must absorb.

Every centimeter of sea-level rise exposes roughly 1.6 million additional people to annual coastal flooding [6], approximately 90% of whom live in low- and middle-income countries [7]. Without additional adaptation, annual flood damages are projected to reach US$14–18T (trillion) per year for 0.9–1.1 m of rise [8]. Estimates of total adaptation costs per unit sea level rise are not available but a lower-bound, rough-order-of-magnitude estimate of US$4T per meter of sea-level rise is reasonable based on a projected cost of US$400B simply to raise dikes by 1 meter along only 3.4% of coastlines [9]. Developing countries currently receive US$21B per year in total adaptation finance across all sectors, not just coastal [10], raising serious questions about the feasibility of coastal adaptation at relevant scales.

Rapidly resolving the uncertainties is essential because effective planning requires reliable sea-level forecasts be delivered before key decisions need to be made. Uncertainty in the rates of sea-level rise compounds uncertainty in inundation thresholds and paralyzes anticipatory spatial planning, routinely defaulting policy toward *in situ* maladaptations that compound sunk costs and strand assets [11], or delaying action until forced by disaster [12]. Unconstrained timelines encourage fragmented, market-based incrementalism that deploys public capital inefficiently [13] and systemically transfers the financial burden of reactive retreat onto vulnerable municipalities and demographics [14].

Current coastal adaptation planning relies heavily on projections from the IPCC Sixth Assessment Report (AR6), which estimates 0.4–0.7 m of rise by 2100 relative to the 1995–2014 baseline [15, 16]. These projections depend on process-based ice sheet models that are forced with climate models and compared against one another in efforts such as ISMIP6. These models, however, are not able to incorporate all of the physics and their choice of initial conditions means causes them to diverge from satellite observations within a decade [17, 18]. As a result, observed GMSL trends (0.66 m median at 2100, 0.41–0.91 m 90% CI) are on track to exceed AR6 medium-confidence projections under all policy-relevant emission scenarios [2, 19]. AR6 documents this in a single figure but without comment: extrapolating the satellite-era rate and acceleration—which reflect the current warming trajectory that could lead to 3 °C (SSP2-4.5)—produces sea-level rise that matches or slightly exceeds the process-based projections under the very high emissions scenario (SSP5-8.5). So while the data show that rates of sea level rise double with approximately 0.4 °C of warming (Fig. 1b), IPCC models need roughly an order of magnitude more warming just to reproduce current observed rates of sea-level rise.

Disagreement between observations and ice sheet models arise from known structural biases, including the absence of leading-order physical processes like iceberg calving [18] and an assumed ice viscosity that is less sensitive to changes in stress than observations indicate [20]. The latter can cause up to ∼30% underestimate of projected 21st century ice loss from West Antarctica, exceeding the uncertainty due to warming trajectories [21]. More broadly, decades of general forecasting research shows that models more complex than their calibration data can constrain produce worse predictions than simpler models [22]. Current ice sheet models have millions of degrees of freedom representing spatially distributed parameters and uncertain parameterizations. The observations used to initialize these models inform a small fraction of these degrees of freedom, with the remainder determined by prior assumptions [23, 24]. A forecasting system based on models representing diverse approaches and a hierarchy of complexity with robust testing against data is needed to fill the needs of coastal planning for sea-level rise and to help guide the development of models that allow for deeper understanding of the salient physical processes.

Data-driven forecasting

We construct a data-driven forecast that focuses on different sea-level contributors, each with minimal physical models constrained by independent observational records that span decades. Observed GMSLR and its sensitivity to observed global mean surface temperature (GMST) are diagnostics, and we show that the sum of our model components agrees within 6% of 2000–2025 observed GMSL in hindcasting tests (Supplementary Material). Unless otherwise noted, all sea-level values in the main text are global-mean 21st-century contributions referenced to the year 2000, while values in the abstract are relative to pre-industrial for consistency with existing projections and literature. Throughout, we label emission scenarios by their approximate end-of-century global mean surface temperature: 2 °C (SSP1-2.6), 3 °C (SSP2-4.5), and 4 °C (SSP3-7.0).

Our framework builds on established lines of data-driven forecasting. The simplest is the "naive" forecast, which requires only a fit to the observed GMSL rate and acceleration. The satellite altimetry record provides GMSLR measurements from 1993–2025 that show a constant acceleration of 0.08 $\mathrm{mm\,yr^{-2}}$ (0.02–0.14 $\mathrm{mm\,yr^{-2}}$) and rate 4.5 $\mathrm{mm\,yr^{-1}}$ (3.5–5.5 $\mathrm{mm\,yr^{-1}}$) in 2025 [2, 19], already exceeding IPCC AR6 projections in 2025 (Fig. 3a). Extrapolating these values to 2100 yields 66 cm (41–91 cm) of 21st-century rise. Assuming the naive forecast has predictive skill over about half the calibration window, the extrapolation of the 32-year satellite record provides predictive skill over the next 15-20 years. Beyond that horizon, the naive forecast is less reliable but nevertheless establishes a scenario-independent, data-driven reference (Fig. 3a).

The next class of data-driven forecasts are the semi-empirical models that fit the observed sensitivity of the rate of GMSLR to changes in GMST (Fig. 1b). Under current warming trends, Rahmstorf (2007) [5] predicts 0.67 m (0.63–0.71 m) of 21st-century rise, matching the naive forecast and the results of this study within error. Although physically motivated by the fit to observed sensitivities, semi-empirical models were assessed as *low confidence* by IPCC AR5 because they do not resolve individual contributing processes and because extrapolation beyond the calibration period assumes a stationary temperature–sea-level relationship [25]. This reasoning is carried forward by AR6 but violated by the decision to favor process models that omit key physical processes, are initialized to snapshots of non-stationary data, and were not tested against observed GMSL, the very quantity they claim to predict, prior to publication [15].

Component forecast model

We decompose global mean sea level into separate components: ocean thermal expansion (thermosteric), Greenland Ice Sheet (GrIS) surface mass balance (SMB) and glacier discharge, West Antarctic Ice Sheet (WAIS) glacier discharge, East Antarctic Ice Sheet (EAIS) SMB, Antarctic Peninsula total ice loss, glaciers (land ice not in the ice sheets), and terrestrial water storage. Our component-summation approach builds on the methodology of previous studies [26–28], while extending to more recent data, initializing the forecasts by blending the calibrated sensitivities with extrapolated observational trends to ensure near-term alignment between the forecast and observed trends (Supplementary Materials), and developing a structured deep-uncertainty treatment of WAIS discharge. The sensitivities of glaciers, GrIS discharge, and Antarctic Peninsula to warming are calibrated with independent observational datasets spanning 24–71 years (Table S1). Surface mass balance for GrIS and EAIS are adopted from published model sensitivities rather than data calibration because they lack sufficient data coverage for our approach [29, 30]. For completeness, we adopt the IPCC projection of terrestrial water storage due to a lack of suitable data to support our approach. Details of our models for EAIS and Antarctic Peninsula can be found in Supplementary Materials; we omit them from the main text because their contributions are small enough to be within the uncertainties of the primary contributors.

The thermosteric component is currently the largest single contributor to observed sea-level rise. We model it with a two-layer energy balance [36] fitted jointly to NOAA *in situ* thermosteric measurements at 0–700 m (1957–2025) and 0–2000 m (2005–2025) [37] (Fig. 2a). In our model, the upper ocean responds to warming GMST through a thermal state variable S_u with relaxation time τ_u. The deep ocean thermal state relaxes toward S_u on a longer timescale $\tau_d = 150\,\mathrm{yr}$ [36]. We fit the model parameters to the data using a Monte Carlo approach described in Supplementary Material. Forecasts are generated by driving the calibrated two-layer model with IPCC AR6 temperature trajectories, propagating posterior parameter uncertainty via Monte Carlo sampling (Supplementary Material).

The Greenland Ice Sheet is currently the largest land-ice contributor due to imbalances between mass gained through snow accumulation and mass loss from melting (SMB) and glacier ice discharge. We treat SMB and ice discharge separately. We find that available data are insufficient to constrain SMB, and instead use regional climate models (RCMs). For the hindcast, we use SMB reanalysis data [31, 38] directly. For projections, we adopt published linear and quadratic sensitivities of SMB to GMST from RCM intercomparisons [29, 30], using the spread across RCMs as structural uncertainty [39] (Supplemental Materials).

GrIS discharge rate is sensitive to ocean temperature [40], and therefore we model it as the cumulative response to time-delayed subsurface ocean warming, with a sensitivity γ, a delay δ between ocean warming and glacier response, and a secular rate r_0 (Supplementary Material). Using 47 years of glacier discharge data [31] and EN4 ocean temperatures [41], we find $\delta = 5$ yr, consistent with fjord overturning and glacier dynamics, with sensitivity $\gamma = 0.37$ $\mathrm{mm\,yr^{-1}\,^{\circ}C^{-1}}$, which is consistent with scaling estimates from ocean and glacier dynamics (Fig. 2b; Supplemental Material). An independently derived discharge record (1986–2023) [33] and NOAA Arctic Report Card estimates [32], both withheld from calibration, shows our model within 90% CI, as does the total Greenland mass balance (Fig. 2c), suggesting that our simple 3-parameter model captures the salient physical processes of the observational record. To project this model forward in time, we calibrate a transfer function between the Greenland regional surface temperature anomaly and the ocean temperature and then apply a normal distribution of Arctic amplification factors to calculate the Greenland regional surface temperature form projected GMST under different emission scenarios (Supplemental Material).

Mountain glacier mass loss is modeled as a rate–temperature relation calibrated against the GlaMBIE global consensus [2000–2023; 42] and GMST. The data only support a linear sensitivity (Supplementary Material), which we find to be $b_{\mathrm{gl}} = 0.66$ $\mathrm{mm\,yr^{-1}\,^{\circ}C^{-1}}$ (Fig. 2d). This is approximately five times the value embedded in IPCC AR6 median projections ($\sim$0.13 $\mathrm{mm\,yr^{-1}\,^{\circ}C^{-1}}$) and 1.3–1.7$\times$ the effective sensitivity of the PyGEM process model [43], but provides excellent fit with global data compilations.

As a validation step, we compare the sum of the independently calibrated components to observations of GMSL from satellite altimetry (Fig. 2e,f). Validation is possible because each component is calibrated against its own observational record. Total GMSL is withheld from calibration. The component sum closes 93–101% of the observed GMSLR rate over three epochs (1993–2025, 2000–2025, 2010–2025), with a variance-weighted attribution assigning the residual mainly to terrestrial water storage (45–56%) and thermosteric (21–27%) components, both within their stated uncertainties (Supplementary Material). The transient sensitivity of our component model, which defines the change in GMSLR rate with warming, closely agrees with observations and the Rahmstorf (2007) [5] semi-empirical results (Fig. 1b).

West Antarctic Ice Sheet.

WAIS is the largest source of uncertainty in sea-level projections (Figs. 2g,h). The ice sheet contains ~3.3 m of sea level equivalent and rests on land that is below sea level and deepens inland, making it susceptible to internal feedbacks commonly called the marine ice sheet instability (MISI; [44]). The delivery of warm, deep seawater is the primary climate driver of mass loss [18]. Ocean thermal forcing sufficient to drive retreat is likely committed throughout the 21st century regardless of emissions pathway [45, 46] and may be driven as much by ocean processes local to the ice sheet as those further afield [47]. Based on these lines of evidence, we make our WAIS projections independent of the warming trajectory, meaning the range of 21st-century outcomes is controlled by ice dynamics that, to leading order, are insensitive to emissions over our forecasting horizon.

No physical process models have been proven to reproduce observations [17] and we find insufficient data coverage and duration to support a data-driven forecast like those described above. To provide a representative range of uncertainties, we construct a scenario-mixture framework grounded in the scientific literature (Fig. 2g,h; Supplementary Material). Scenario 1: 'slow WAIS' where rates of discharge continue to increase within the range of observed values and with linear relationship between SLR contributions and ocean temperatures. This scenario admits 25–85 mm GMSLR contribution by 2100. Based on the changes in WAIS discharge over the past decades and the position by the IPCC AR6, we take this scenario to be unlikely, and assign it a 10% probability. Scenario 2 represents the role of MISI-driven acceleration and grounding-line retreat, yielding a GMSL contribution of 0.15–1 m by 2100. The lower value is anchored to transient-calibrated ice sheet model projections [48] and the observed 33 km grounding-line retreat at Pine Island Glacier since 1992 [49] while the upper bound is set by structured expert judgment [50]. We assign Scenario 2 an 80% probability. Scenario 3 represents MISI combined with the potential for cascading structural failure of ice cliffs, known as the marine ice cliff instability (MICI), with the range defined by the IPCC AR6 low-confidence 83rd (0.6 m) and 95th (1.3 m) percentile ranges. We assign a 10% probability to this scenario, consistent with IPCC AR6 'deep uncertainty' framing. Within each scenario, the 2100 endpoint is drawn from a skew-normal distribution in log-space with physically motivated asymmetry [51]. A multiplicative rheology correction (R with median $R = 1.3$) is applied to all scenarios, reflecting the underestimation of ice loss when models assume the ice-viscosity stress exponent $n = 3$ rather than the observationally constrained value $n = 4.1 \pm 0.4$ [20, 21, 52]. The resulting mixture has a median 21st century sea-level rise of 0.41 m with a 0.07 to 1.56 m 90% CI at 2100. We refer to this mixed model as 'fast WAIS' because it allows for, but does not force, rapid ice mass loss (Scenarios 2 and 3). Sensitivity tests of the scenario weights show the chosen weights do not qualitatively impact the findings of this study (Supplementary Materials).

Forecasts

We drive the calibrated model with the IPCC AR6 warming trajectories of 2 °C (SSP1-2.6) through 4 °C (SSP3-7.0), the pathways most consistent with current policies and stated pledges (Fig. 3). To ensure our sea-level trajectories are faithful to the observed trends in the first decade of the forecasting window, we smoothly blend extrapolations of observed trends for the next 10 years with the first 10 years of the forecasts so that the first year (2026) of our forecast is almost exclusively the extrapolated GMSLR rates from satellite observation, the forecast in 2031 has an even weighting between our component model and the naive forecast, and by 2036 the forecast is taken entirely from the calibrated components forced by different emission scenarios and warming trajectories.

For the 2 °C target, our forecast shows a median of 0.79 m of sea-level rise [0.46–1.89 m] relative to year 2000, which is 72% above the IPCC AR6 medium-confidence estimate of 0.46 m. Under 3 °C warming, approximately the current trajectory, the median is 0.96 m [0.62–2.07 m], 66% above IPCC's 0.58 m. For the 4 °C trajectory, the median reaches 1.16 m [0.82–2.26 m]. For reference, the 5 °C warming trajectory (SSP5-8.5) yields a median of 1.35 m [0.99–2.45 m].

Relative to pre-industrial GMSL (adding ∼0.19 m of observed rise through 2000), these forecasts correspond to medians of 0.98–1.35 m of global mean sea level rise across the 2-4 °C policy-relevant pathways. The probability of exceeding 1 m relative to pre-industrial by 2100 is 47% under 2 °C warming, 76% under 3 °C, and 96% under 4 °C warming trajectory. The probability of exceeding 2 m relative to pre-industrial by 2100 is approximately 10%, with a minimum of 6% if warming is held below 2 °C. Under the condition that WAIS loses ice slowly, the probability of exceeding 1 m of sea level rise relative to pre-industrial levels effectively drops to zero for less than 3 °C of warming, implying lower flood risks for tens of millions of people and reduced costs of adaptation and damages of order trillions of US dollars (Fig. 3b). As validation, our component forecast for 3 °C of warming closely agreement with extrapolations of observed GMSLR trends, as should be expected if the warming trend remains similar to the current trend and WAIS, which has contributed only ∼4% to GMSLR to date, continues to lose mass at its current rate.

Across the range of possible WAIS mass loss and emission scenarios, our forecasts suggest that WAIS is likely to become the dominant source of sea-level rise before the end of the century (Fig. 4a). It is uncertain when that will happen and depends on how quickly emissions decrease. The stack of mean values shows the relative contributions of the primary sea-level contributors, showing that the thermosteric component likely will remain the single largest contributor in the 21st century unless WAIS mass losses over take it. Greenland and mountain glaciers will continue to lose mass and contribute to sea level rise throughout the 21st century.

The distribution of the uncertainties shows a dramatic separation between the predictable and, currently, unpredictable components (Fig. 4b). These results show that WAIS accounts for virtually all of the variance, and approximately 92% of the confidence interval, in 21st century sea-level rise. The other sea level contributors, all of which have calibrated sensitivities to global mean surface temperature, contribute approximately 1% to the total variance and 8% to the 90% (CI) uncertainty range across the plausible warming scenarios. As a result, the 90% uncertainty range of 1.47 m for 3 °C is four times larger than the spread across warming scenario medians (0.37 m from 2–4 °C). This picture is fundamentally different under the condition that WAIS loses mass slowly. In the slow-WAIS scenario, the warming-sensitive components become the largest contributors to the uncertainty (righthand column of Fig. 4b). The relative magnitude of uncertainties reveals that sea-level risk decomposes into two approximately orthogonal dimensions, each addressed by a distinct class of societal response.

Orthogonal dimensions of sea-level risk

The first risk dimension is sensitive to emissions reduction. The temperature-dependent components are sensitive to changes in global mean surface temperature. In this work, we calibrate the sensitivities using decades of observations. Their collective contribution shifts systematically across warming trajectories and is reducible by emissions cuts (Fig. 4a). Under the 'slow WAIS' scenario, the forecast narrows to a 90% range of only ∼14 cm, an order of magnitude narrower than the full scenario cast that allows for rapid mass loss from WAIS due to internal feedbacks (Fig. 3). In the slow scenario, emissions policy is the primary lever for managing risk: the difference between 2 °C and 4 °C warming is roughly 20 cm of additional rise, which roughly translates to 30 million additional people exposed to coastal flood risks and US$50B/yr in coastal adaptation costs.

The second risk dimension is relatively insensitive to emissions reductions. WAIS dynamics accounts for virtually all of the total forecast variance and is largely decoupled from the emission pathway. The tail of the forecast distribution—the outcomes that drive the most severe human and economic consequences—is controlled almost entirely by whether WAIS retreat remains gradual at modern rates of loss or transitions to rapid, irreversible retreat. Emissions reduction decreases the median but does not substantively change the tail of the distribution. The probability of exceeding 2 m relative to pre-industrial is 6% under the 2 °C trajectory and 14% under 4 °C, with a 10% probability across emission scenarios. The 6% lower bound on this range highlights the fact that emission reductions alone cannot retire the high risk outcomes.

The risks and costs to coastal communities helps to highlight the dichotomy between the two possible scenarios of fast vs slow WAIS mass loss, or equivalently, emissions-insensitive vs emissions sensitive (Figs. 3 and 4a). The difference at the 5% exceedance probability, one possible design threshold for coastal adaptation that we denote by stars in Fig. 3b, is 1.4 m of sea-level rise by 2100 under the current policy warming trajectory (3 °C). This difference translates into roughly 260M more people exposed to annual flood risks, US182$B/yrincoastaladaptation, andUS$18.2T/yr (∼16% global GDP per year) in coastal flood damages without further adaptation. These large numbers reflect global population densities, with the highest densities near coastlines, and the fact that the locations of coastal communities and infrastructure are calibrated to lower sea levels. For intuition, the fast-WAIS scenario at 5% exceedance implies an order of magnitude more sea level rise in the next 75 years than was experienced in the past 125 years, while the slow-WAIS scenario implies a factor of 3.5 more rise.

Resolving the question of WAIS mass loss rates is among the most financially consequential goals in modern science (Fig. 4c). A value-of-information analysis shows that perfect knowledge of whether WAIS will lose mass slowly or rapidly is worth of order US$5T in avoided costs, with the value peaking when confidence in a slow-WAIS future reaches 80% and the information arrives early enough to inform infrastructure decisions. The value decays with time because late-arriving information leaves fewer years of decisions to improve, and because all forecasts, including this study, degrade as they extend beyond their calibration bounds. These numbers frame the research investment question in concrete terms: even partial reduction of WAIS uncertainty, well short of perfect knowledge, would justify annual research expenditures orders of magnitude above current levels. The return on that investment is determined by how quickly the observations can narrow the range of WAIS outcomes, which depends on both the rate of data collection and the development of reliable process models, like those used in the IPCC, capable of providing explanatory insights that underscore deep understanding of ice sheets and climate.

Our forecast identifies a single measurement that would do more to reduce sea-level uncertainty than any other: the rate of West Antarctic ice loss over the next two decades. If WAIS mass loss continues near current rates through the 2030s and 2040s, that would build the case for the slow-WAIS scenario and allow the forecast to narrow toward the predictable, lower, emissions-sensitive range where adaptation costs are more manageable. If instead grounding-line retreat accelerates beyond the pace of our central scenario, the distribution would shift toward higher outcomes and 1.5–2 m of rise would move from a tail risk to a planning baseline. Either outcome would reshape coastal adaptation worldwide and would justify investment well beyond current levels. The observations needed to distinguish these futures can be made within a generation, but only if the observing systems and research programs are sustained and expanded at a scale that reflects the costs at stake.

Scientific research is a necessary condition for providing actionable information for coastal adaptation, but it is as likely as not that further research will add to the growing evidence to support rapid WAIS mass loss rather than build confidence in a slow-WAIS future [18]. Sustaining a wide range of uncertainties that admit $< 1m$ and $> 2m$ of GMSLR in 2100 at 10% probabilities, will force decisions on coastal planning, adaptation, and managed retreat that impact millions of lives and cost billions of dollars to be made under large uncertainties. The extent of these uncertainties in terms of lives, property, and ecosystems raises the question: Can we take action to increase the potential stability of West Antarctica? How much harm to people and the environment can we prevent? Recently, some research directions have been proposed to answer this question [53] with initial models of glacier interventions aimed at stabilizing WAIS show promise [54]. As with any nascent research, there are many unresolved questions about the feasibility, unintended consequences, and governance of these interventions [55]. There are risks in intervening to remediate the West Antarctic Ice Sheet and slow its loss and the logistical challenges will undoubtedly be great, but these risks and challenges must be weighed against the risks and challenges of sea-level rise as described in this study. There is no zero risk, zero cost response to sea level rise. The costs and challenges are real and must be borne regardless of what decisions are made, the question is how much cost, how much challenge, and how much harm. The answer to that question hinges largely on how fast West Antarctica will lose ice, and nothing short of a virtual guarantee in slow WAIS mass loss provides comfort for the world's coastlines.

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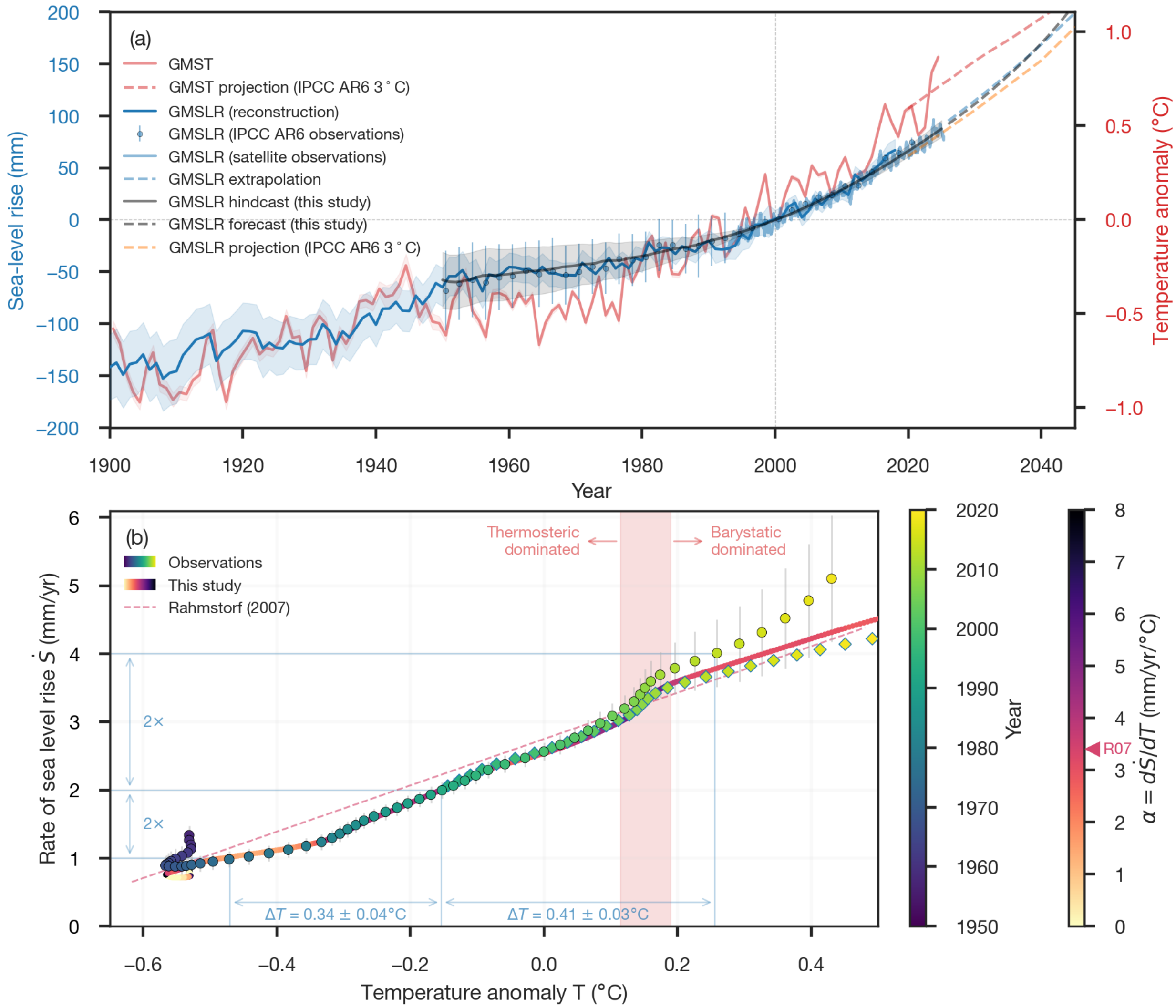

Figure 1: **Observed global mean sea-level rises with global mean surface temperature.** **(a)** GMSLR from reconstructions and tide-gauge records (blue line [1]; blue circles [15]) along with more recent NASA satellite altimetry (light blue; [4]) show sea-levels rising with warming. Hindcast results of this study (grey line with 90% CI shading) show good agreement between our model and previously published observations and models. The hindcast starts at 1950, the earliest year when GMSL reconstructions agree [15]. Different projection methods (dashed lines) show the divergence of IPCC AR6 projections and observed trends within a decade. GMST (red; [3]) is referenced to the right hand axis. **(b)** Rate of GMSLR ($\dot{S} = dGMSLR/dt$) versus GMST anomaly, T. Observed values are shown in circles [1] and diamonds [4], colored by year, and the component model of this study is a line colored by instantaneous transient sensitivity $\alpha = d\dot{S}/dT$. The Rahmstorf (2007) semi-empirical model is shown in a color corresponding to its α value [5]. The vertical red band indicates the shift in the primary contributor from ocean thermal expansion changes in glacier ice and water storage (barystatic). The sinuous tail in the early observations is largely due to dam impoundment of water rather than a thermodynamic signal [1].

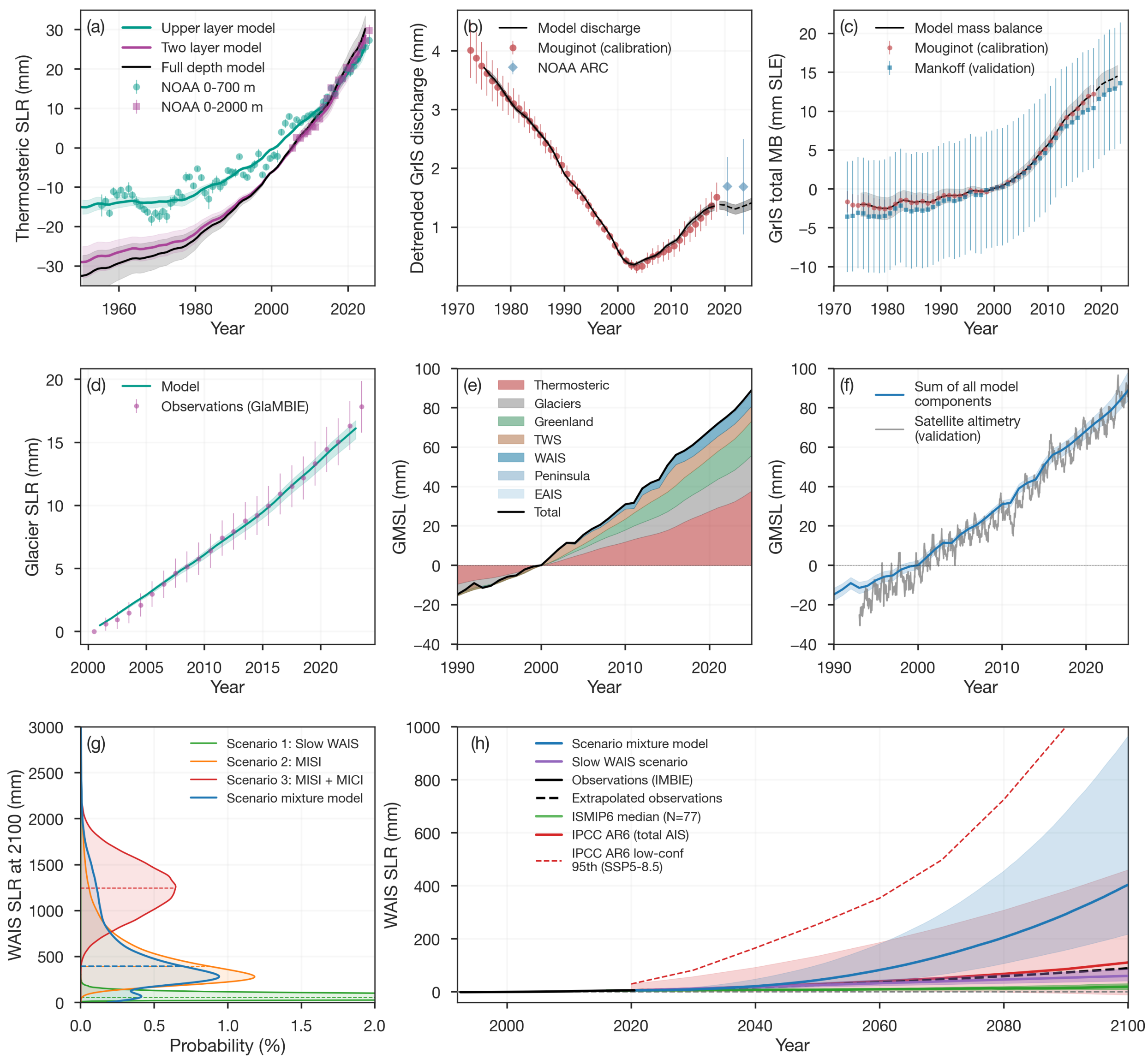

Figure 2: **Model calibration, validation, and WAIS uncertainty.** **(a)** Our two-layer energy balance model fitted jointly to NOAA thermosteric values at different depth layers with the full depth model used in the forecasts (Supplementary Materials). **(b)** Greenland ice discharge modeled as a cumulative time-delay response to subsurface ocean warming with the calibration data from [31] shown with filled circles and diamonds showing NOAA Arctic Report Card estimates [32]. **(c)** Greenland total mass balance with validation data shown in blue squares with 90% CI error bars [33]. **(d)** Global glacier model fitted to GlaMBIE consensus observations (2000–2023) using a linear temperature-driven model. **(e, f)** Budget closure: temperature-forced component hindcast (stacked) compared with satellite altimetry data (cf. Fig. 1). **(g)** WAIS scenario distributions at year 2100. **(h)**WAIS projections from our full scenario mixture (blue) and slow WAIS scenario (purple) compared with observations (black, [34]), the ISMIP6 ensemble (green; [35]), and IPCC AR6 projections (red).

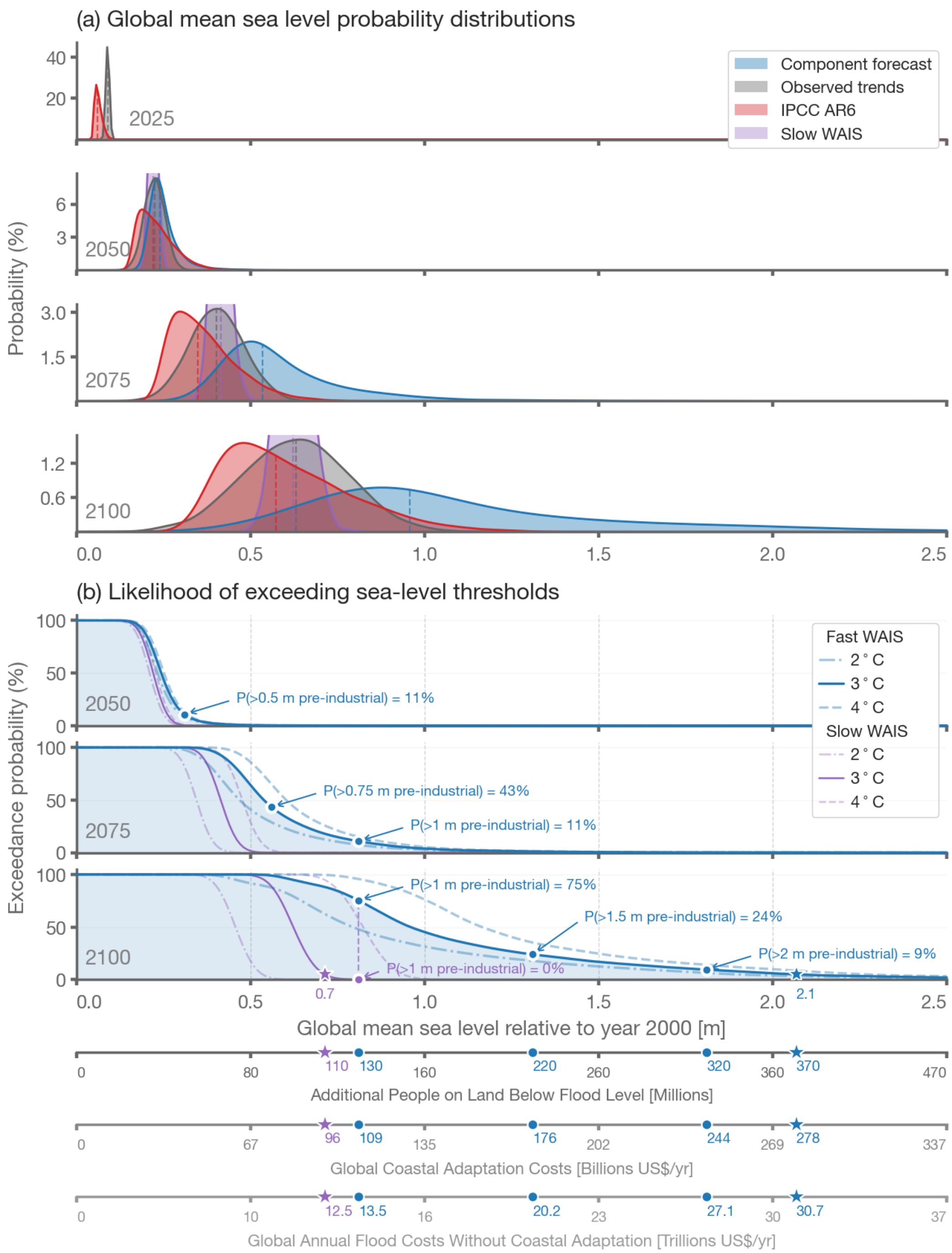

Figure 3: **Global mean sea-level probabilities.** (a) Probability distributions for different forecasting approaches and the IPCC projections. The 'slow WAIS' scenario (purple, clipped for clarity) assumes sustained mass loss at approximately modern rates (Scenario 1). Vertical dashed lines indicate the respective medians. (b) Probability of exceeding a given sea-level threshold for different warming trajectories including the WAIS scenario mixture model that allows for fast mass loss (blue) and the slow WAIS scenario (purple). Lower axes show estimates of the corresponding number of people currently living on land that will be below flood level [6] and financial costs [8, 9] (Supplementary Material). Stars indicate the 5% exceedance probability.

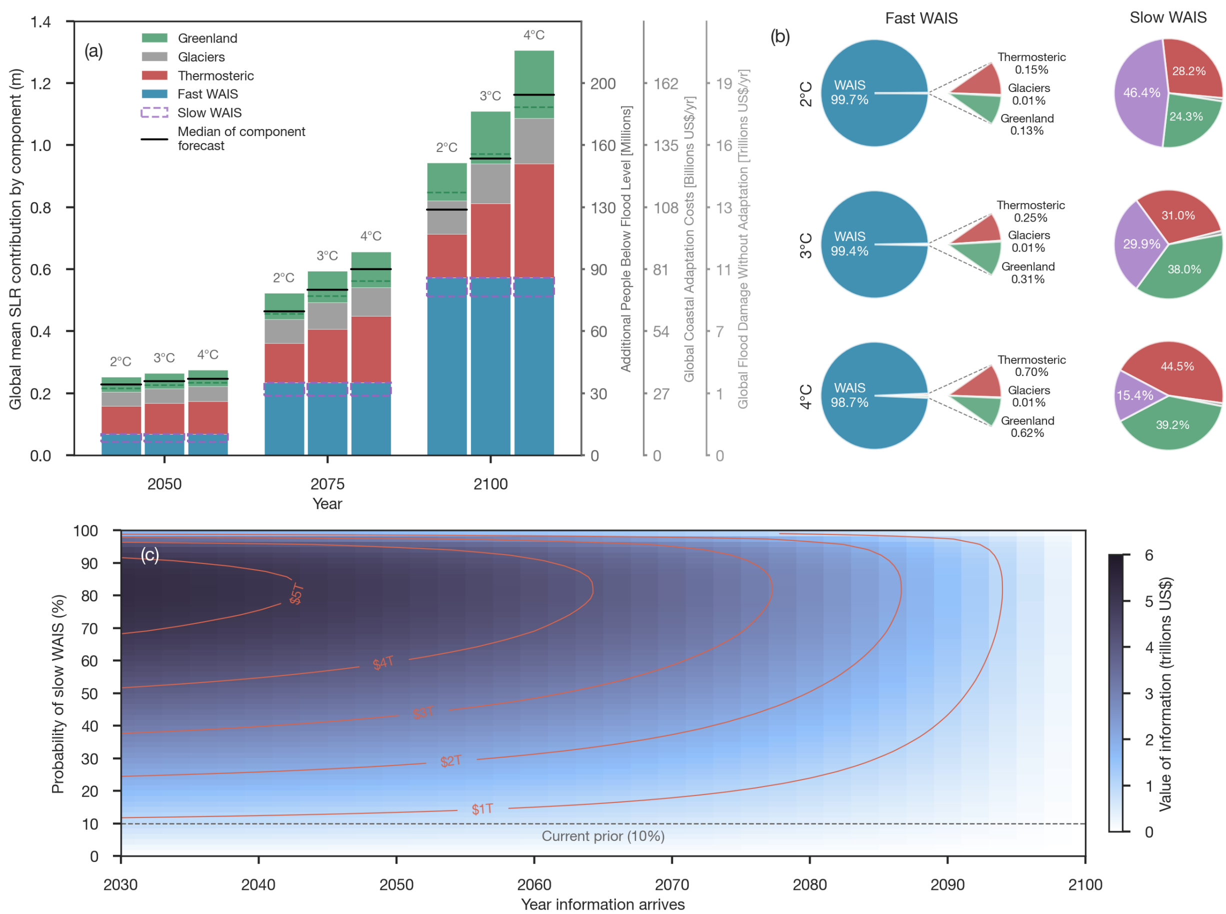

Figure 4: **Component contributions, variance decomposition, and value of information. (a)** 21st century mean GMSLR contribution by component and respective societal costs. Green dashed lines within Greenland bars partition discharge anomaly (below) from SMB anomaly (above). The purple box shows how much WAIS will contribute if it loses mass slowly while the rest of the blue bar shows the potential SLR from rapid ice loss. **(b)** Variance-fractions at 2100. WAIS accounts for 97–99% of total forecast variance when rapid ice loss is possible. Fan breakouts show the non-WAIS contributions. If WAIS loses mass slowly, the variance distributions fundamentally shift (right column) **(c)** Financial value of resolving WAIS uncertainty as a function of when information arrives. Contours and colormap in trillions US$ based on coastal adaptation costs (Supplemental Material). The plot shows that information has more value if it arrives early enough to inform key decisions and if it provides sufficient confidence that WAIS will lose mass slowly, allowing planning around lower future sea levels.